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CONTINUOUS ASSESSMENT 1
II YEAR – SEMESTER 4
CSE23AE204 - PCP III
(B.Tech. E01,E02,E03,E05,E06)(B.Sc E61)

SET F

Question 1: Problem Statement

A customer support analytics system analyzes **customer chat messages** to detect **symmetric emotional patterns**.

A message is considered **symmetric** if a **contiguous part of the message reads the same forwards and backwards** (palindrome).

Given a string message, your task is to identify and return the **longest symmetric substring** present in the message.

If multiple substrings of maximum length exist, return **any one of them**.

Example 1

Input:

message = "babad"

Output:

"bab"

Input Format

- A single string message representing a customer chat message.

Output Format

- A string representing the **longest palindromic substring** found in the message.

Question 2: Problem Statement

Problem Statement

You are given a lowercase English string and a non-negative integer k.

Each character in the string is converted to its **alphabetical position** (a = 1, b = 2, ..., z = 26).

All converted values are concatenated to form a numeric string.

You must then perform the following operation **exactly k times**:

- Replace the current number with the **sum of its digits**

Return the final value after completing all k iterations.

Input Format

- A lowercase string s
- An integer k

Output Format

- Return an integer representing the final computed value

Example 1

Input:

s = "abc"

k = 1

Output:

6

Explanation:

Character conversion:

- a → 1
- b → 2

Question 3: Problem Statement

You are developing a **premium calculation module** for an insurance company. The system processes **one insurance policy at a time**, provided as a **space-separated string**. Each policy belongs to a specific category and follows different premium calculation rules.

The system must:

- Identify the policy type from input
- Calculate the final premium amount
- Use **abstraction and polymorphism** to support multiple policy types
- Assign a unique policy ID using a **static variable**
- Return the calculated premium

Example 1

Input

```
{ "policy": "HEALTH Ravi 5000 1500" }
```

Output

6500.0

Explanation

The health policy premium is calculated by adding the age-based risk amount to the base premium.

Premium Rules

- A base premium is charged to all the policy.

Health Insurance Policy

- An age-based risk amount is added.
- The total premium is the sum of base premium and risk amount.

Vehicle Insurance Policy

- A percentage-based vehicle value charge is added.
- The total premium is the sum of base premium and value-based charge.

Life Insurance Policy

- A fixed coverage amount is added.
- The total premium is the sum of base premium and coverage amount.

Input Format

A single **space-separated string**.

Health Policy: HEALTH <CustomerName> <BasePremium> <AgeRiskAmount>

Vehicle Policy: VEHICLE <CustomerName> <BasePremium> <VehicleValuePercentage>

Life Policy: LIFE <CustomerName> <BasePremium> <CoverageAmount>

Output Format

Return a **double value** representing the **final insurance premium**.